Application Architecture Patterns

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Patterns

don't reinvent

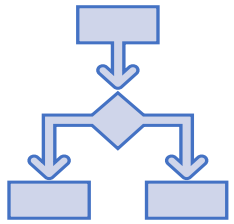
THE WHEEL

Layered

Multi-Tier or “Three Tiered”

Presentation Tier

- Mobile Application
- Web Application



Logic Tier

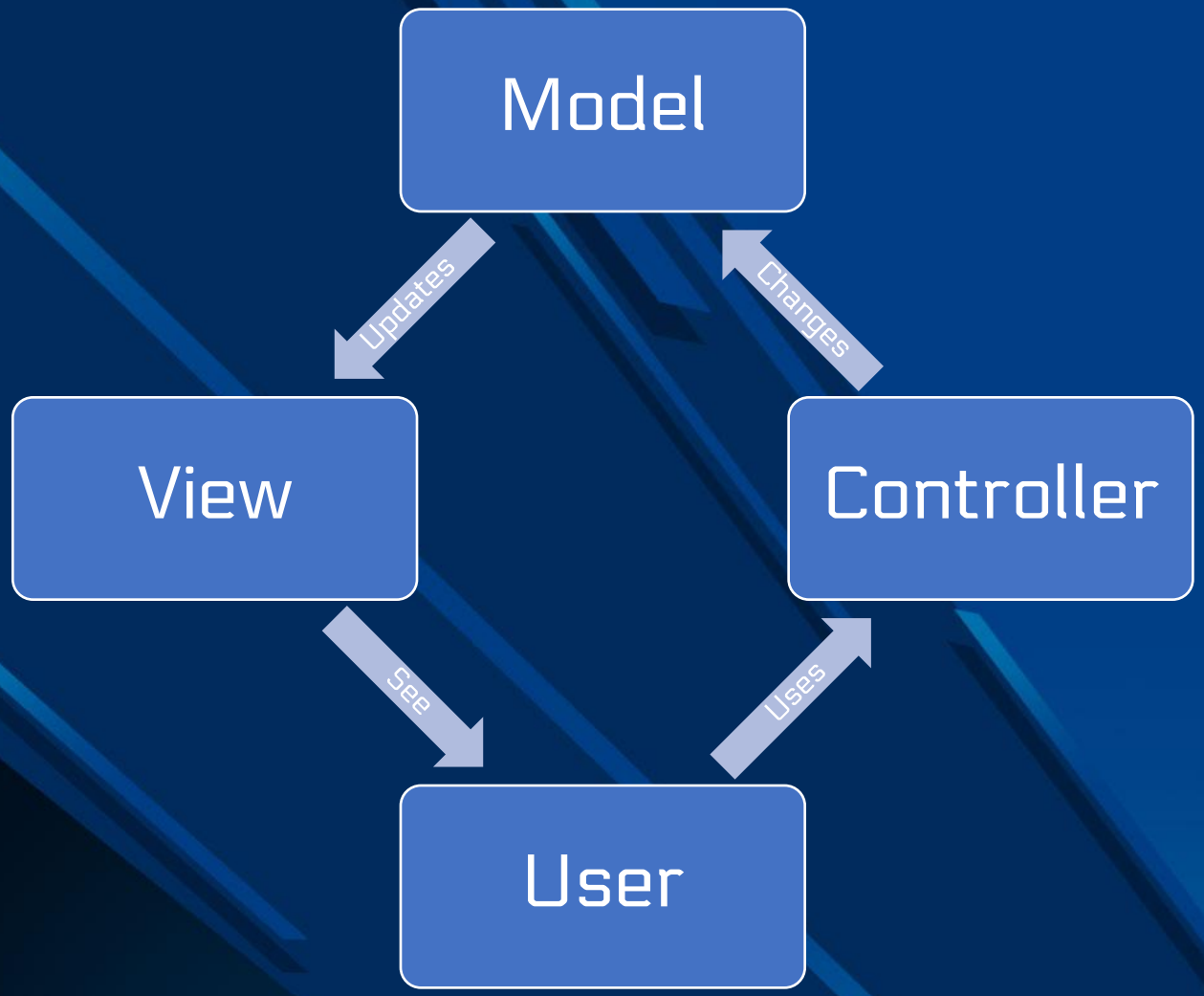
- Place Order
- Ship Order

Data Tier

- Database
- Storage

Model, View, *

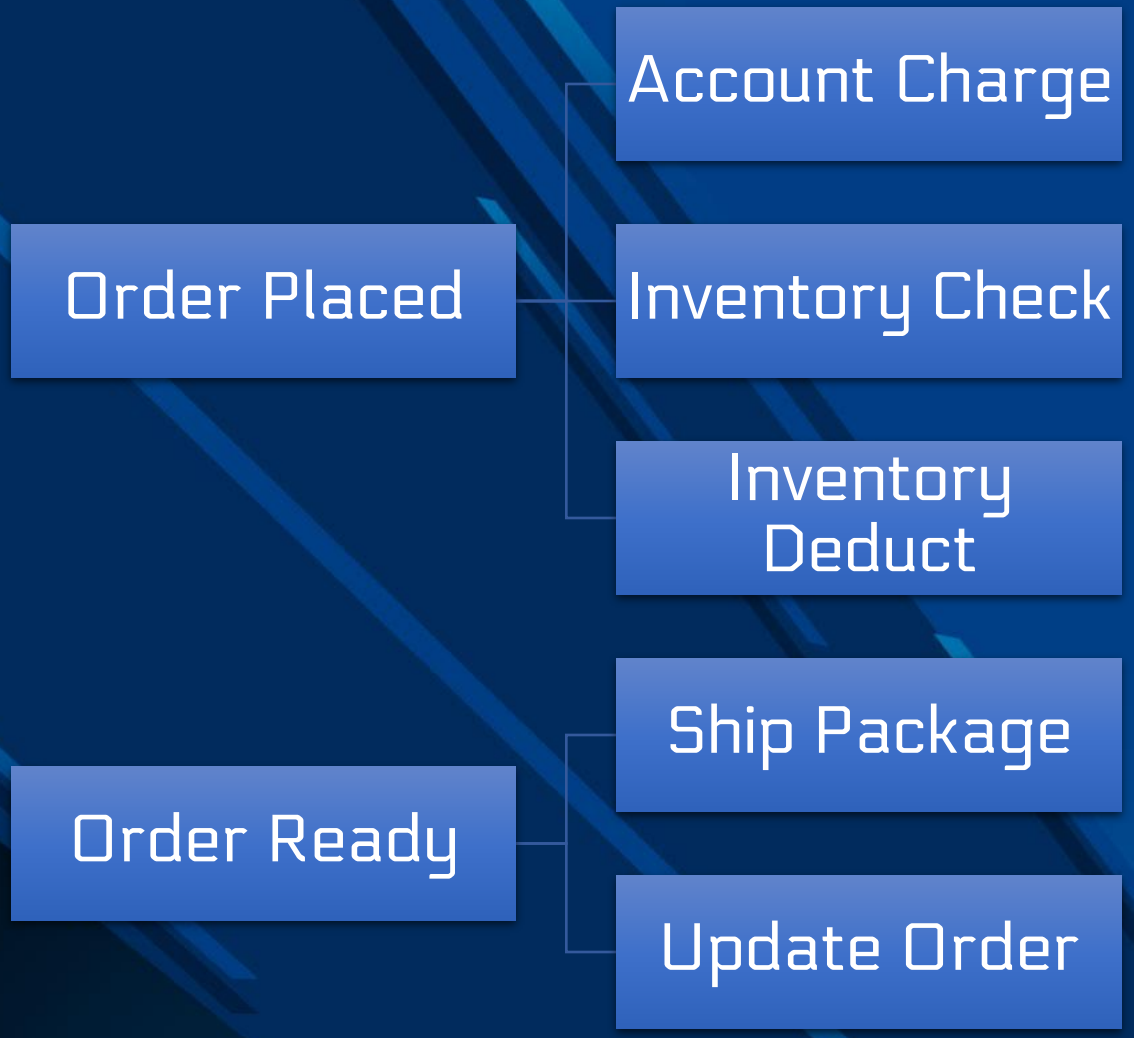
- MODEL – VIEW – CONTROLLER (MVC)
- MODEL-VIEW-PRESENTER (MVP)
- MODEL-VIEW-VIEW MODEL (MVVM)



Event Driven

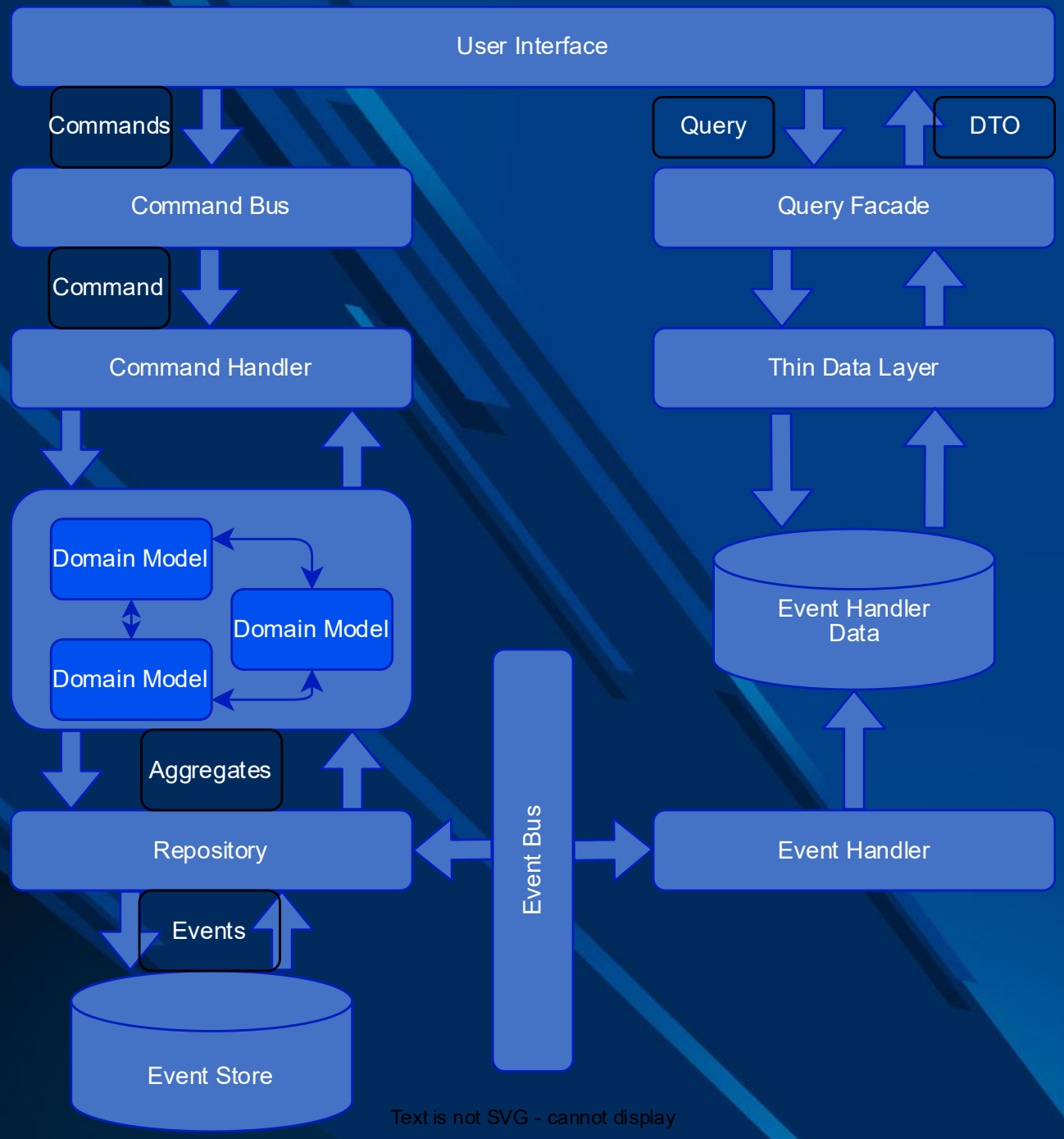
Event Sourcing

Publish/Subscribe



CQRS

COMMAND & QUERY RESPONSIBILITY SEGREGATION



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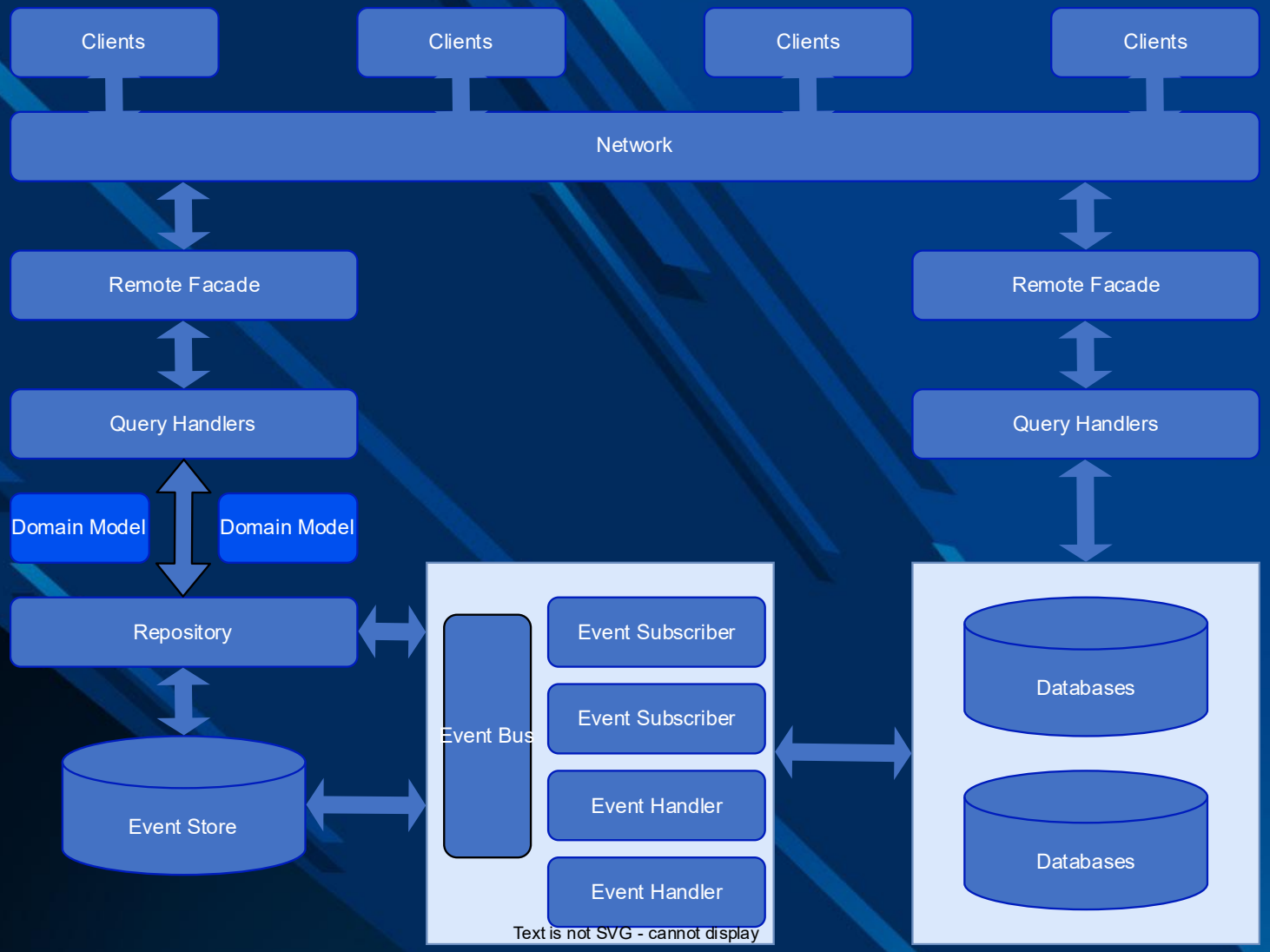
CQRS + Event Sourcing



Plug-In

Micro-Kernel

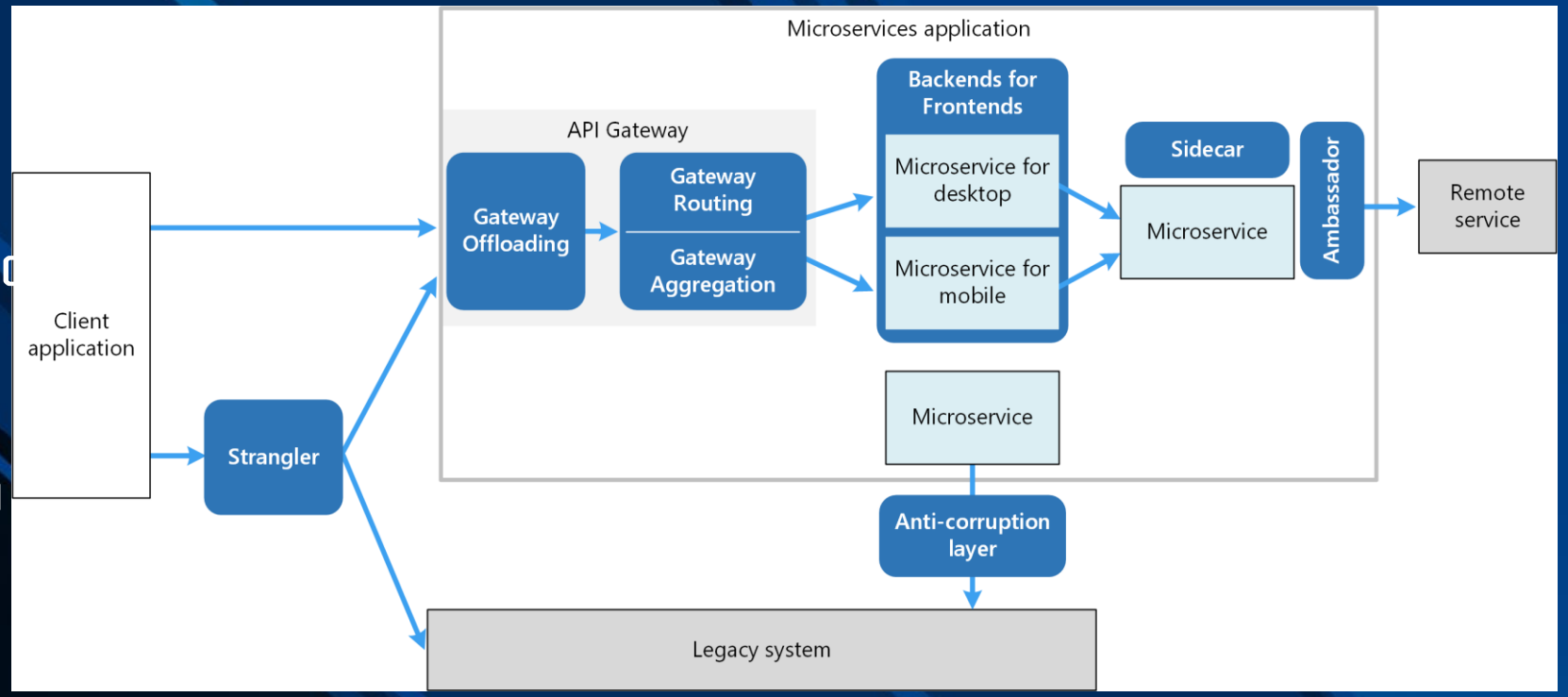
Service Oriented

SOA

Microservices

Common Microservice Pattern

- Ambassador
- Anti-corruption layer
- Backends for Frontend (BFF)
- Bulkhead
- Gateway Aggregation
- Gateway Offloading
- Gateway Routing
- Sidecar
- Strangler Fig



Serverless

The “Cloud”

Not on my PC

Toolbox

Q & A